TPs sur les firmwares

1. TP1: Analyse statique avec binwalk

J'ai choisi un firmware de chez DLink: https://ftp.dlink.de/dir/dir-605l/driver_software/DIR-605L_fw_reva_116b01_ALL_de_20150508.zip

J'ai effectué les commandes suivantes afin d'extraire le firmware.

```
unzip DIR-605L_fw_reva_116b01_ALL_de_20150508.zip
binwalk -Me 2025.03.03-12.36_DIR_825I_RT8197G_WW_1.0.5_release.bin
```

Résultats:

```
mathieu@[illegible] in ~/Documents/IoT/NETGEAR as [illegible] took 0s
↳ binwalk -Me 2025.03.03-12.36_DIR_825I_RT8197G_WW_1.0.5_release.bin
[2026-04-01T09:01:31Z ERROR binwalk::extractors::common] Failed to execute command sasquatch["/home/mathieu/Documents/IoT/NETGEAR/extractions/2025.03.03-12.36_DIR_825I_RT8197G_WW_1.0.5_release.bin.extracted/280000/squashfs_280000.sqsh"]: No such file or directory (os error 2)
[2026-04-01T09:01:31Z ERROR binwalk::extractors::common] Failed to spawn external extractor for 'squashfs' signature: No such file or directory (os error 2)
[2026-04-01T09:01:31Z ERROR binwalk::extractors::common] Failed to execute command sasquatch["/home/mathieu/Documents/IoT/NETGEAR/extractions/2025.03.03-12.36_DIR_825I_RT8197G_WW_1.0.5_release.bin.extracted/280000/squashfs_280000.sqsh"]: No such file or directory (os error 2)
[2026-04-01T09:01:31Z ERROR binwalk::extractors::common] Failed to spawn external extractor for 'squashfs' signature: No such file or directory (os error 2)

                                                
```

Avec cet offset, nous allons extraire tout le bloc de données concernant le fs SquashFS via `dd` :

```
dd if=2025.03.03-12.36_DIR_825I_RT8197G_WW_1.0.5_release.bin bs=1 skip=2621440
```

Ce fichier créé "filesystem.sqsh", est à présent exploitable afin d'entrer directement dans le FS du système. Nous allons utiliser squashfs-tools:

```
sudo unsquashfs filesystem.sqsh
```

Le résultat est le suivant:

```
┌─mathieu@karlmarx in ~/Documents/IoT/NETGEAR as 🍲 took 0s
└─λ sudo unsquashfs filesystem.sqsh
Parallel unsquashfs: Using 8 processors
2117 inodes (1844 blocks) to write

[=====================================================================

created 1716 files
created 442 directories
created 268 symlinks
created 133 devices
created 0 fifos
created 0 sockets
created 0 hardlinks
```

Et le contenu du dossier est le FS complet:

```
┌─mathieu@karlmarx in ~/Documents/IoT/NETGEAR/squashfs-root🔒 as 🍲 took 0s
[●] × tree -L 1
.
├── bin
├── boot
├── dev
├── etc
├── home
├── lib
├── lib32 → lib
├── lib64 → lib
├── mnt → /tmp/mnt
├── opt
├── proc
├── root
├── sbin
├── srv
├── storage
├── sys
├── system
├── tmp
├── usr
├── var → /tmp/var
└── VERSION

19 directories, 3 files
```

Quelques fichiers sont à remarquer:

```
┌─mathieu@karlmarx in ~/Documents/IoT/NETGEAR/squashfs-root🔒 as 🕵 took 0s
└─λ cat etc/passwd etc/shadow
File: etc/passwd
root:x:0:0::/root:/bin/sh
nobody:x:65534:65534::/tmp:/bin/false
dmsd:x:1000:1::/:/bin/false
dsysinit:x:1001:1::/:/bin/false
dwatcher:x:1002:1::/:/bin/false
scheduler:x:1003:1::/:/bin/false
linkwatcher:x:1004:1::/:/bin/false
mfc:x:1005:1::/:/bin/false
tr:x:1006:1::/:/bin/false
easymesh:x:1007:1::/:/bin/false
quagga:x:1008:1005:Quagga priv drop user:/:/bin/false

File: etc/shadow
root:*:::::::
nobody:*:::::::
dmsd:*:::::::
dsysinit:*:::::::
dwatcher:*:::::::
scheduler:*:::::::
linkwatcher:*:::::::
mfc:*:::::::
tr:*:::::::
easymesh:*:::::::
quagga:*:::::::
```